MINISTÉRIO DA CIÊNCIA E TECNOLOGIA
INSTITUTO NACIONAL DE PESQUISAS ESPACIAIS

The logic of arguments

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Houston, we have a problem

Pesquisa FAPESP

The language barrier

Study suggests correlation between researchers' productivity and their competence in writing in English

"It's hard to find good science in poorly written papers"
Joan W. Bennett (editor, Mycology)

"All articles written in bad English are rejected in the peer review process because the knowledge cannot be understood at the level required for publication."
Robert McMeeking (editor, Journal of Applied Mechanics)

Writing well in English is not enough...

Pesquisa
FAPESP

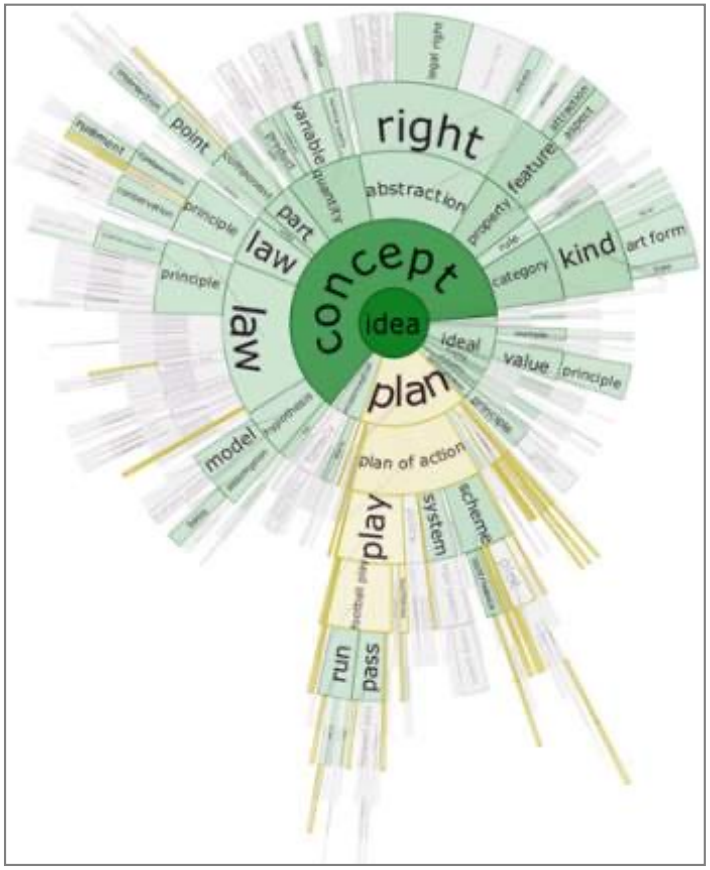
"I am an editor of a scientific journal. It is common to receive articles written by researchers who did not know how to plan their research or execute the experiments and failed to adopt a methodology. The fact that they also do not know how to write in English is a detail in a chain of problems."

Benedito Barraviera (Brazilian Association of Scientific Editors)

What's the problem with bad articles?

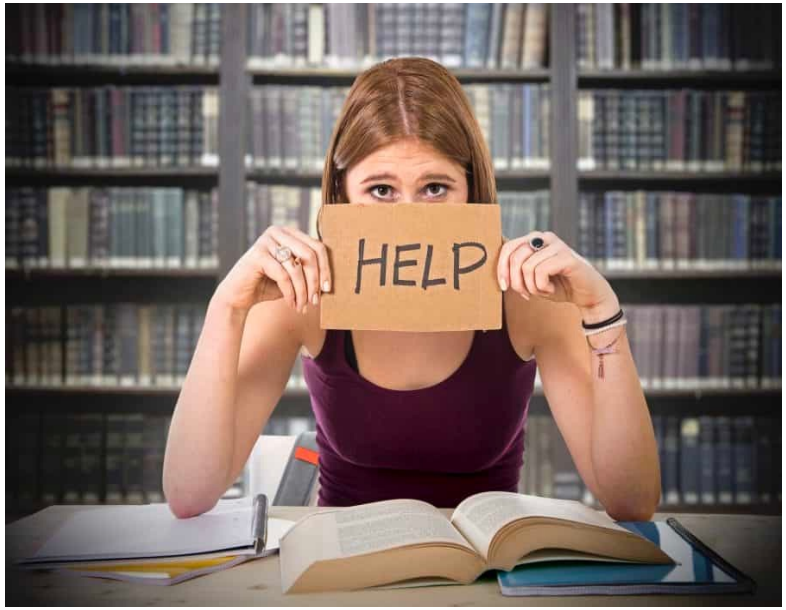
Pesquisa
FAPESP

"An exercise that I always propose to my students is to detect the paper's question. Many fail to do this. A lot of scientific work is just repeating something that has been done."

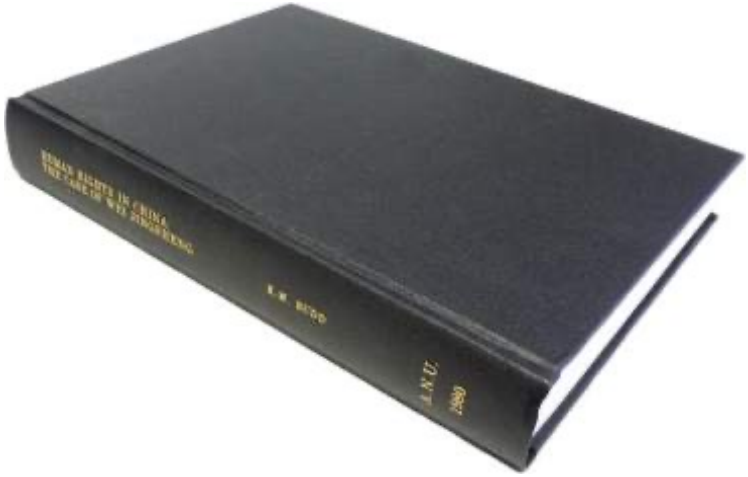


"Brazilian researchers lack knowledge and training to understand what a scientific article is and what kind of approach is tailored for publication in international journals"
Márcia Triunfol (former assistant editor of Science)

Thesis Development



Submit scientific papers with results as soon as possible



Compose the thesis from the papers

Spiral writing

Planning

↓

Generation of the first version

↓

Technical reviews (content)

↓

Style revisions (clarity and readability)

↓

Peer and advisor reviews

1. Introduction

What is the scientific problem?

What is the motivation of the thesis?

Where does the problem come from?

What am I going to do?

2. Hypothesis + Methods

How do you plan to solve/approach the scientific problem?

What is different from the literature?

What will be your contribution to the advancement of knowledge?

3. Results (case studies)

What were the main results of the study?

What is the statistical validity and uncertainty?

What does the reader need to know?

4. Discussion

What do your results show?

How do they compare with the literature?

What is the power of generality of the results?

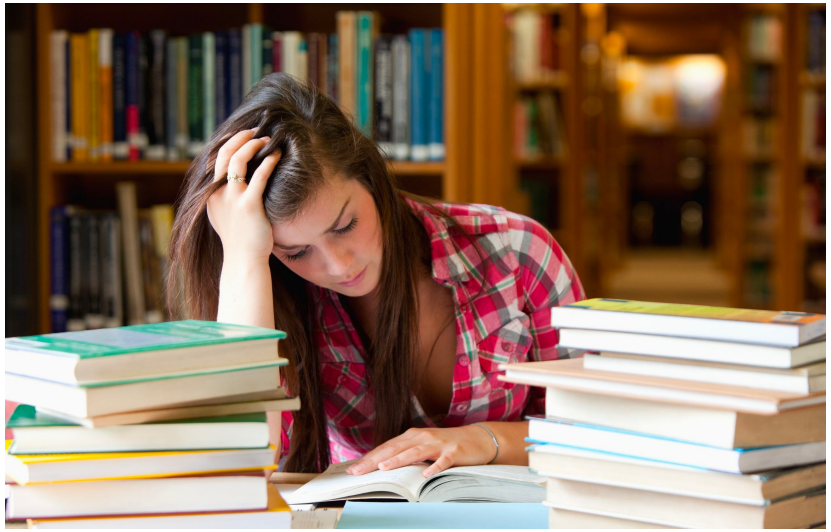
What is the effective contribution to the advancement of knowledge?

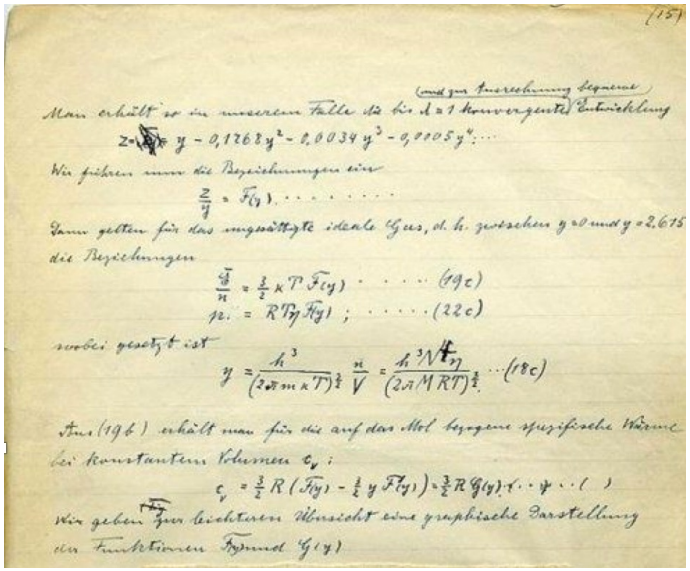
Section	Question to be answered	Purpose	Expected length with A4 paper, font size 10–12 and 1·5 line spacing
Introduction	Why did you start?	Summarise the context of your study and state the aims clearly	1 page
Methods	What did you do?	Give enough detail for the study to be repeated	2–3 pages
Results	What did you find?	Describe the study sample and use the data analyses to answer the aims	2–3 pages
Tables and figures	What do the results show?	Clarify the results	3–6 tables or figures
Discussion	What does it mean?	Interpret your findings in context of other literature and describe their potential impact on health care	2–3 pages
References	Who else has done important work in your field?	Cite the most relevant and most recent literature	20–35 references
Total document			**12–20 pages**

fonte: "Scientific Writing: easy when you know how" (Peat et al.)

You'll be judged by your arguments

The importance of a good argument





Good arguments are essential to convince readers

How to convince your readers?

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Arguments and Evidence

Argument ⟷ Evidence

Arguments are what we want the reader to agree with

Evidences are the experiments and results obtained by the authors

Arguments e Evidences

Argument ⟷ Evidence

In experimental or synthetic science, we need to provide evidence that allows comparison between different arguments (e.g., performance of algorithms, field samples, statistical assessment)

Arguments and Evidences

Argument ⟷ Evidence

A: This machine learning algorithm was able to discriminate different types of land cover

E: This has been verified using reliable field samples and compared to previous ML methods

Arguments and Evidences

Argument ⟷ Evidence

A: This algorithm can efficiently place labels on maps

E: This was verified by comparing the performance of the algorithm with others in the literature using a standardized test suite

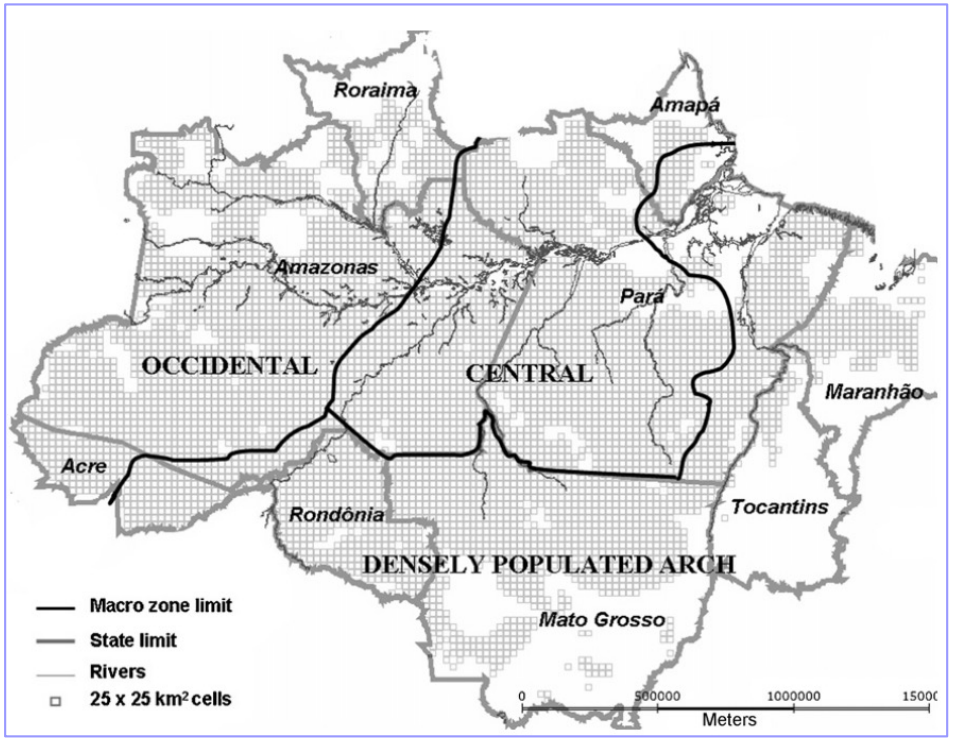
Arguments and Evidences

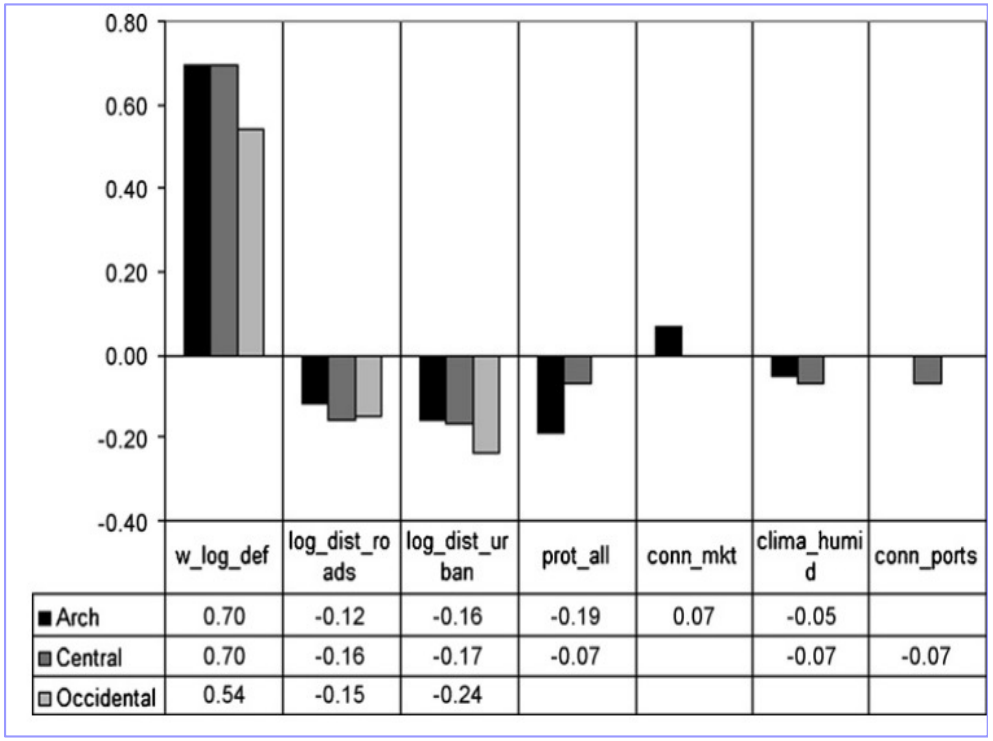
Argument ⟷ Evidence

A: There are intra-regional differences between factors associated to deforestation in Amazonia

E: This was verified by building spatial regression models for three spatial partitions of Amazonia

Arguments and Evidences

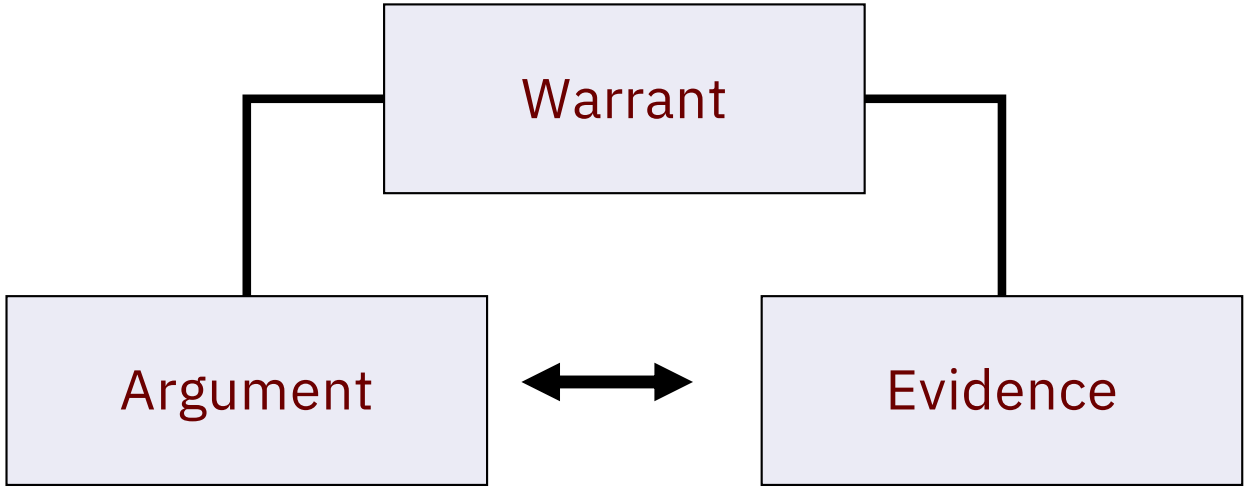




	w_log_def	log_dist_roads	log_dist_urban	prot_all	conn_mkt	clima_humid	conn_ports
Arch	0.70	-0.12	-0.16	-0.19	0.07	-0.05	
Central	0.70	-0.16	-0.17	-0.07		-0.07	-0.07
Occidental	0.54	-0.15	-0.24				

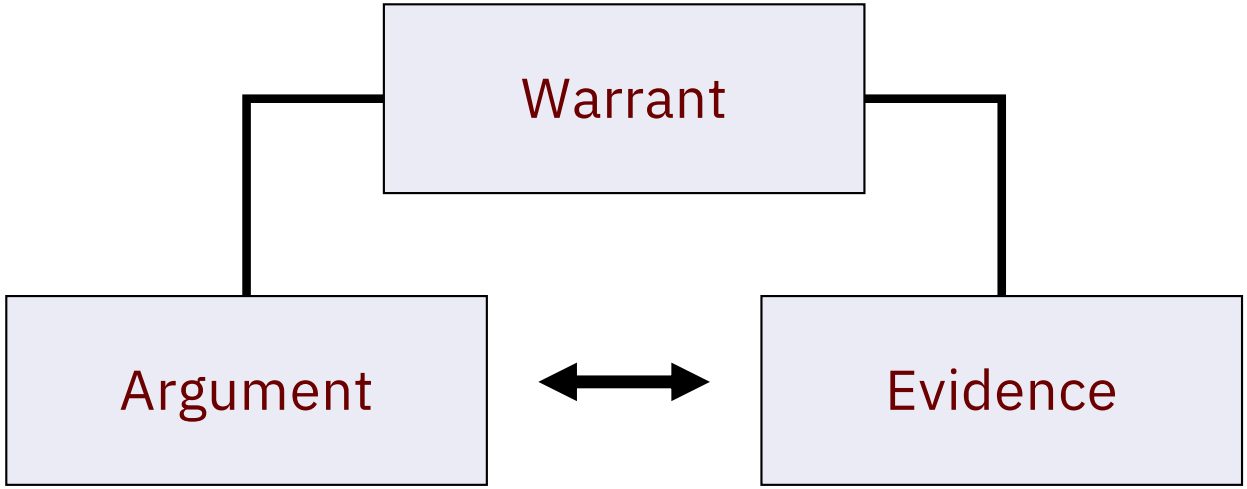
Spatial statistical analysis of land-use determinants in Brazilian Amazonia: Exploring intra-regional heterogeneity (Aguiar et al., 2007)

Warrants



1. A warrant is an established principle that connects the arguments and the evidence
2. Based on scientific consensus

Warrants



A: This image interpretation procedure was able to discriminate different stages of soybean production

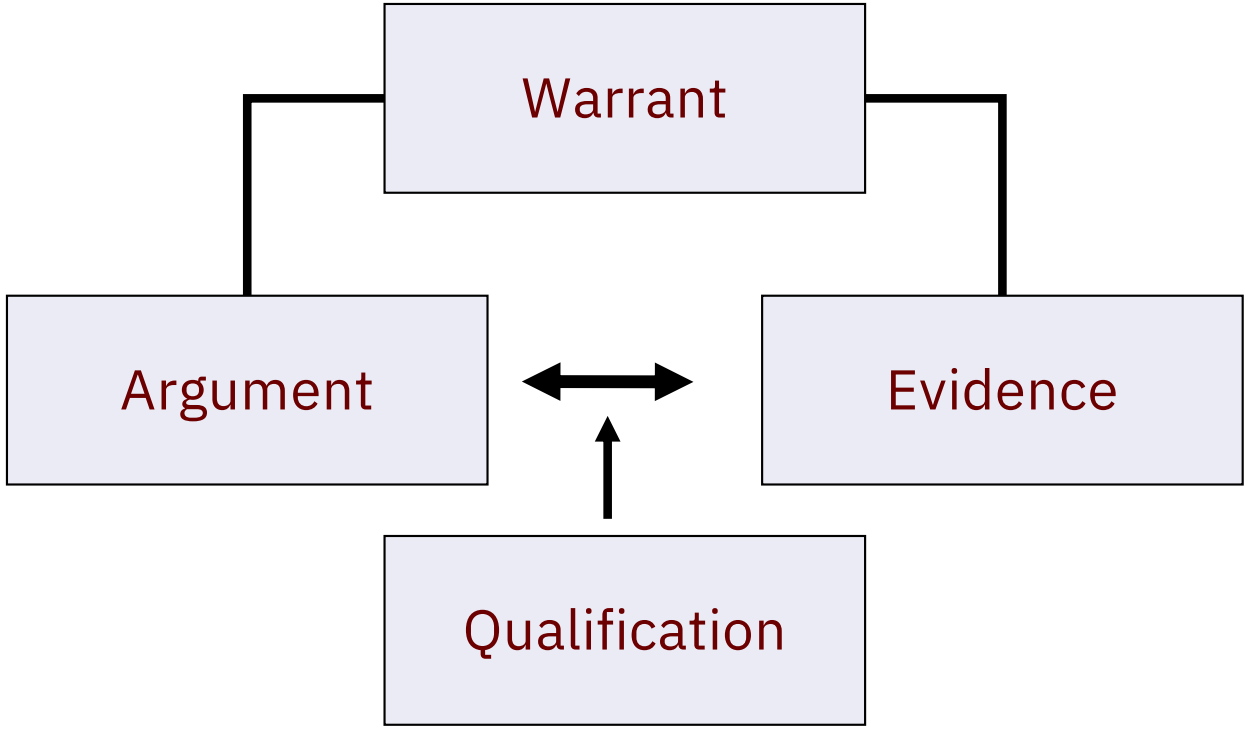
E: This was verified with a field survey with well-constructed sampling

G: Field surveys, done at the time of image acquisition using established statistical techniques, are an accepted way to verify remote sensing estimates.

Qualifications

1. Qualifications limit the scope of your conclusions
2. Stipulate the conditions under which your arguments are valid
3. Respond to potential objections from readers
4. Show that you are an accurate and careful author

The Role of Qualifications



Qualifications indicate the power of generality of the result

Qualifications on Computer Science papers

The development of analytical software for big EO data faces several challenges.... Arguably, it is unlikely that a single approach will emerge as the complete best solution for big EO analytics.

In the design of `sits`, we had to make choices...

The limitations of `sits` should also be considered.

("Satellite Image Time Series Analysis for Big Earth Observation Data", Simões et al., 2021)

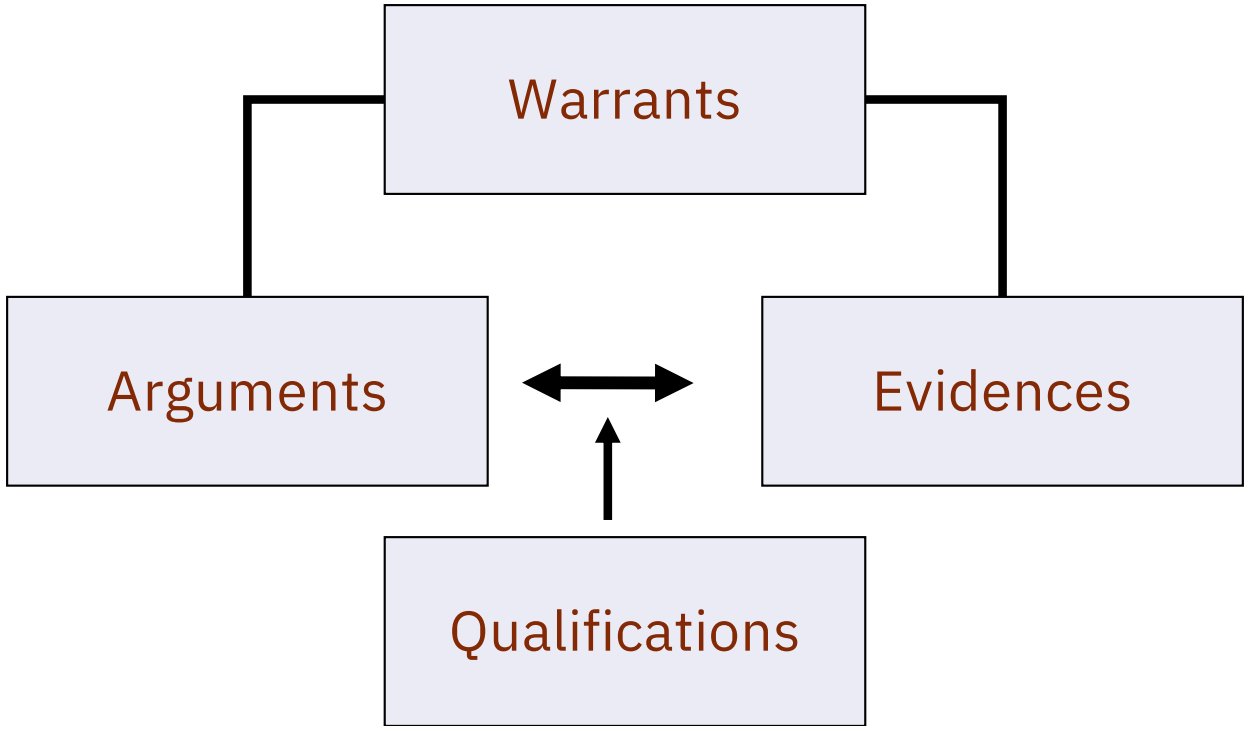
Arguments, evidence, guarantees and qualifications

Argument – your contribution to answering a scientific question

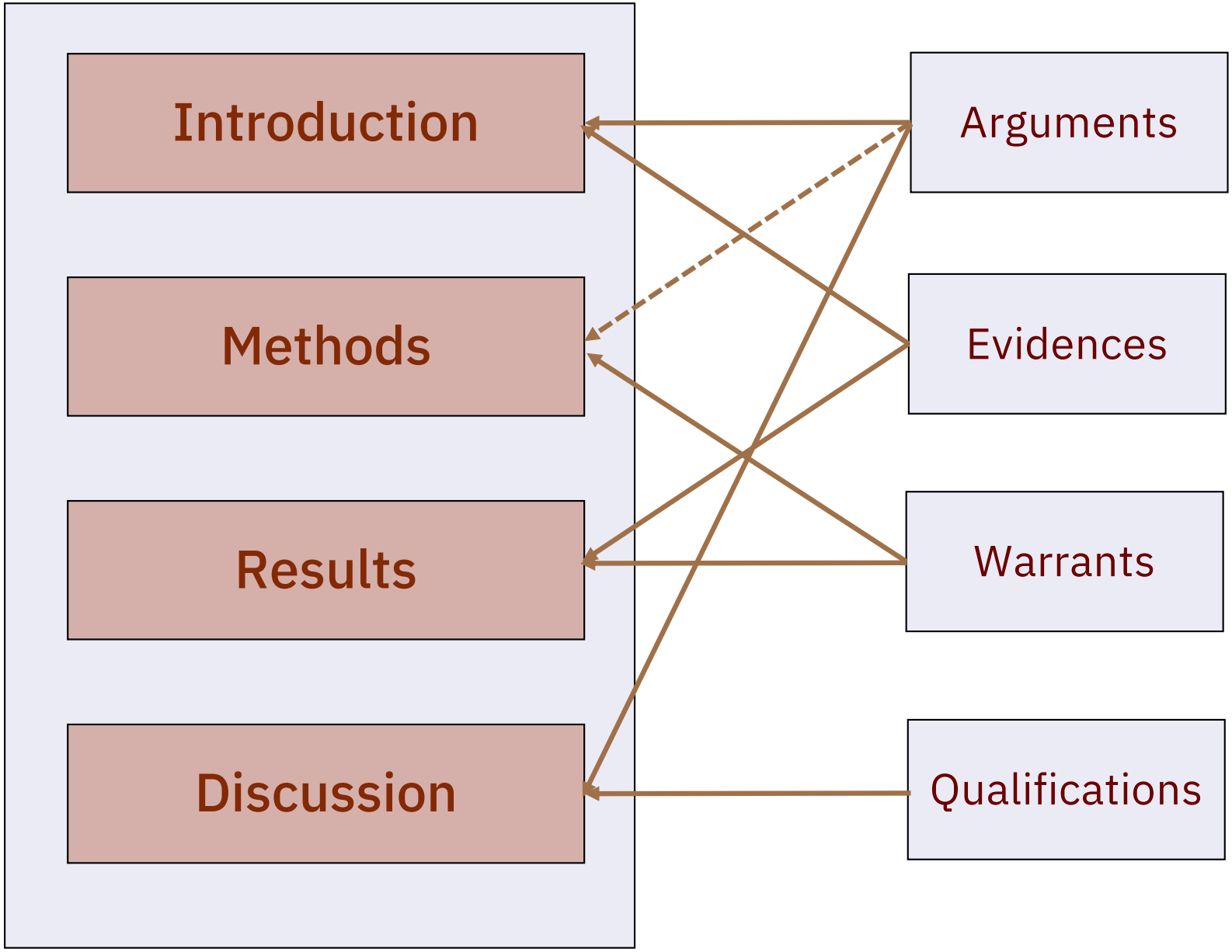
Evidence – proof of your contribution

Warrants – the basis of your work

Qualifications – limits of your arguments



IMRAD and the logic of arguments



Example of argumentation

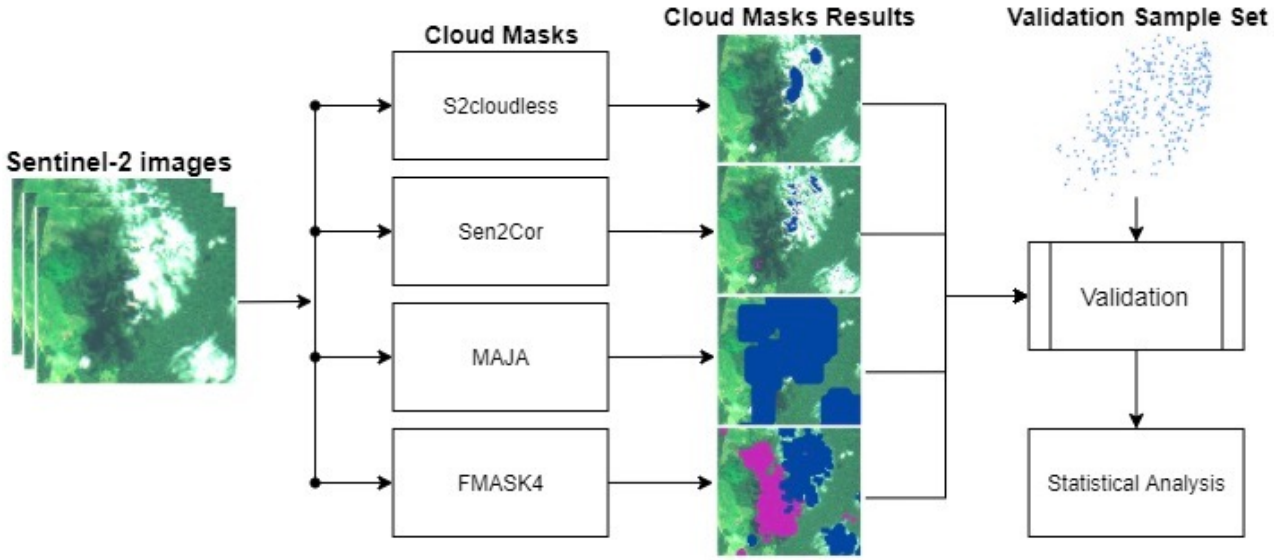
remote sensing

MDPI

Article

Comparison of Cloud Cover Detection Algorithms on Sentinel–2 Images of the Amazon Tropical Forest

Alber Hamersson Sanchez [1,*,†], Michelle Cristina A. Picoli [2,†], Gilberto Camara [2,†], Pedro Ribeiro Andrade [1,†], Michel Eustaquio D. Chaves [3,†], Sarah Lechler [4], Anderson R. Soares [2], Rennan F. B. Marujo [2], Rolf Ezequiel O. Simões [2], Karine R. Ferreira [2] and Gilberto R. Queiroz [2]



Arguments, evidence, warrants and qualifications

Abstract: Using more than eight thousand validation points, we compared four cloud detection methods: Fmask 4, MAJA, Sen2Cor, and s2cloudless. Our results point out that FMask 4 has the best overall accuracy on images of the Amazon region (90%), followed by Sen2Cor (79%), MAJA (69%), and S2cloudless (52%) (Argument).

Introduction: Cloud detection algorithms are an active research field [2-8]. Each algorithm has specific characteristics and ad hoc techniques; thus, comparing them on a theoretical basis is hard. In practice, performance assessment is done by selecting representative images and assessing how well each algorithm performs in each image. Baetens et al. compare three cloud detection methods (MAJA, Sen2Cor, and Fmask) using 32 images from 10 different locations. (Argument + Evidence + Warrant).

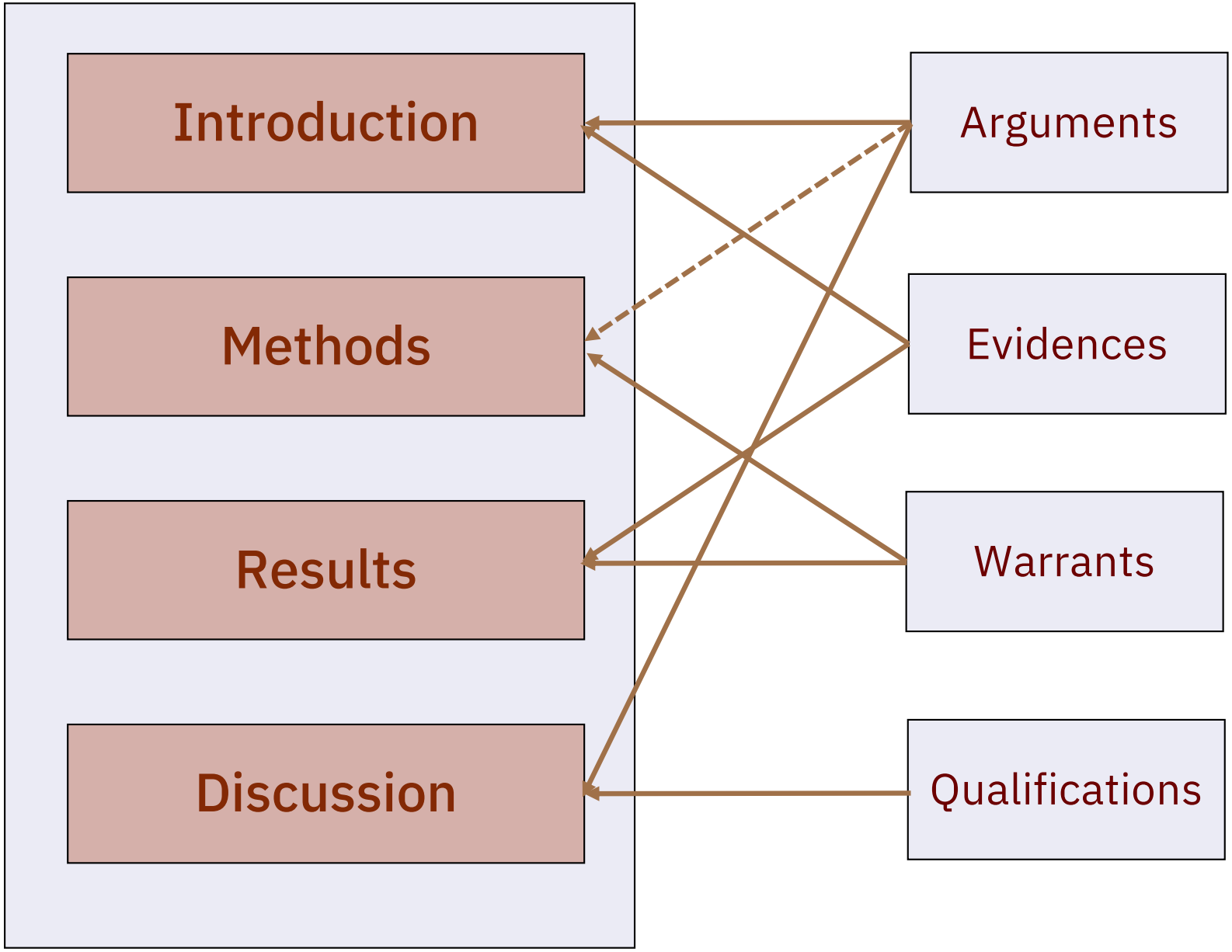
Arguments, evidence, warrants and qualifications

Methods: To validate the resulting cloud masks, we used sample points tagged by remote sensing experts through visual interpretation, following the work of Foga et al. and Zhu et al. We selected a set of random locations inside each Sentinel--2 tile. Since Sentinel--2 images at 10 m resolution have over 120 million pixels, standard statistical techniques indicate that using about 400 samples per image is enough to achieve a 95% confidence level with a 5% margin of error (Israel,1992). (Argument + Evidence + Warrant)

Results: In our experiments, Fmask 4 has the best overall accuracy, followed by Sen2Cor, MAJA, and S2cloudless (see Table 4). (*Argument + Evidence*).

Arguments, evidence, warrants and qualifications

Discussion: The results of this study show that the Fmask 4 algorithm consistently performs better than the alternatives for Sentinel-2 images of the Amazon rain forest. Our results are different from those of Baetens et al. We now consider some hypothesis that could account for such significant differences. *(Argument + Qualification)*

The comparison done by Baetens et al. uses 10 different sites, including equatorial forests, deserts and semi-deserts, agricultural areas, mountains, and snowy areas. (...) Our study deals only with forest and agriculture areas; the images tested have no deserts, mountains or snow. By focusing on the Amazon biome, our results are intended as guidance for experts interested in measuring land change in the region. Given its focus, these results cannot be generalised to non-forest regions. *(Argument + Qualification)*

IMRAD and the logic of arguments